

# Computer Science & IT



## Compiler Design



Parsers  
Lecture No. 04



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# Topics to be Covered



Topic

LL(1)

Topic

Topic



|                                                                                                                                                            | First                             | Follow                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----------------------------------|
| $\underline{E} \rightarrow (\underline{T} \cdot \underline{E'})$<br>$\underline{E'} \rightarrow + \underline{T} \underline{E'} / \underline{\epsilon}$     | $\{id, ( \}$<br>$\{+, \epsilon\}$ | $\{\$, )\}$<br>$\{\$, )\}$       |
| $\underline{T} \rightarrow \underline{F} \underline{T'}$<br>$\underline{T'} \rightarrow \underline{*} \underline{F} \underline{T'} / \underline{\epsilon}$ | $\{id, ( \}$<br>$\{*, \epsilon\}$ | $\{+, \$, )\}$<br>$\{+, \$, )\}$ |
| $\underline{F} \rightarrow \underline{id} / (\underline{E})$                                                                                               | $\{id, ( \}$                      | $\{*, +, \$, )\}$                |



|    | id                  | +                         | *                     | (                   | )                         | \$                        |
|----|---------------------|---------------------------|-----------------------|---------------------|---------------------------|---------------------------|
| E  | $E \rightarrow TE'$ |                           |                       | $E \rightarrow TE'$ |                           |                           |
| E' |                     | $E' \rightarrow +TE'$     |                       |                     | $E' \rightarrow \epsilon$ | $E' \rightarrow \epsilon$ |
| T  | $T \rightarrow FT'$ |                           |                       | $T \rightarrow FT'$ |                           |                           |
| T' |                     | $T' \rightarrow \epsilon$ | $T' \rightarrow *FT'$ |                     | $T' \rightarrow \epsilon$ | $T' \rightarrow \epsilon$ |
| F  | $F \rightarrow id$  |                           |                       | $F \rightarrow (E)$ |                           |                           |

LL(1)



|                                                        | first             | Follow      |
|--------------------------------------------------------|-------------------|-------------|
| $S \rightarrow \underline{(S)} / \underline{\epsilon}$ | $\{(, \epsilon\}$ | $\{), \$\}$ |

$($   $($   $)$   $)$   $\$$   
 $\uparrow$   $\uparrow$   $\uparrow$   $\uparrow$   $\uparrow$

|     | $($                                   | $)$                                        | $\$$                     |
|-----|---------------------------------------|--------------------------------------------|--------------------------|
| $S$ | <u><math>S \rightarrow (S)</math></u> | <u><math>S \rightarrow \epsilon</math></u> | $S \rightarrow \epsilon$ |

accept

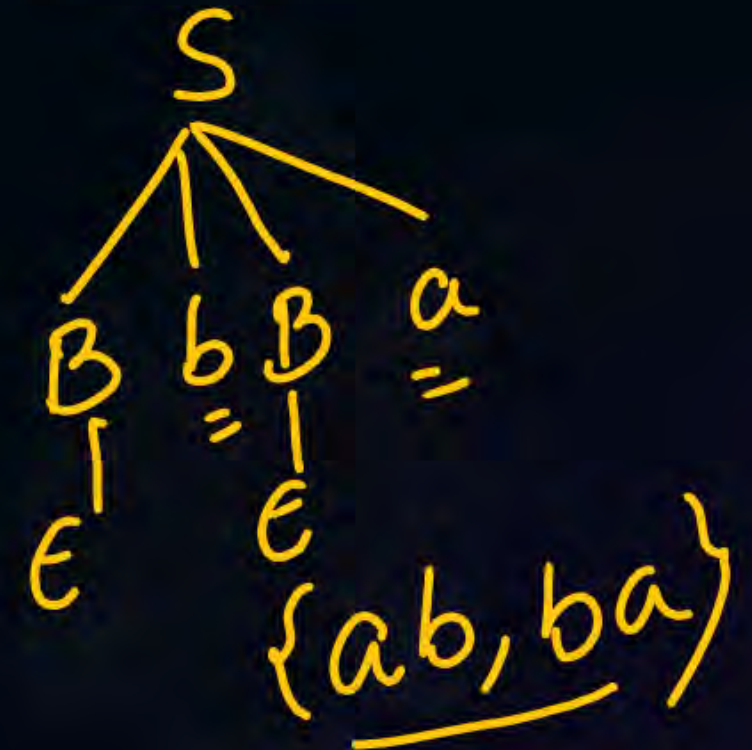
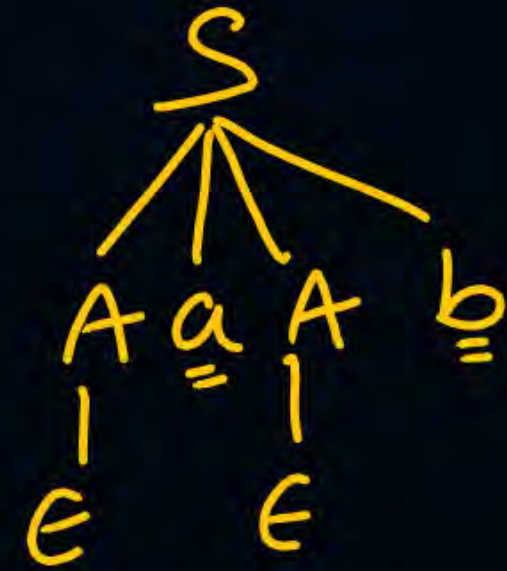
|                             |
|-----------------------------|
| <del><math>(</math></del>   |
| <del><math>S</math></del>   |
| <del><math>)</math></del>   |
| <del><math>(</math></del>   |
| <del><math>S</math></del>   |
| <del><math>)</math></del>   |
| <del><math>S</math></del>   |
| <del><math>\\$</math></del> |



|                                                     | first()        | follow()   |
|-----------------------------------------------------|----------------|------------|
| $S \rightarrow \underline{AaAb} / \underline{BbBa}$ | $\{a, b\}$     | $\{\$ \}$  |
| $A \rightarrow \epsilon$                            | $\{\epsilon\}$ | $\{a, b\}$ |
| $B \rightarrow \epsilon$                            | $\{\epsilon\}$ | $\{a, b\}$ |

LL(1) parsing table

|   | a                        | b                        | \$ |
|---|--------------------------|--------------------------|----|
| S | $S \rightarrow AaAb$     | $S \rightarrow BbBa$     |    |
| A | $A \rightarrow \epsilon$ | $A \rightarrow \epsilon$ |    |
| B | $B \rightarrow \epsilon$ | $B \rightarrow \epsilon$ |    |





## LL(1) parsing algo:-

Suppose 'X' is the Symbol on top of the stack

and 'a' is current I/p Symbol

- 1) If  $x = a = \$$ , then Successful Completion
- 2) If  $(x == a) \neq \$$ , pop 'x' and advance the i/p pointer
- 3) If  $\underline{x}$  is a non terminal, then take the production  $\underline{m[x, a]} = \{ \underline{x} \rightarrow \underline{uvw} \}$  then replace  $\underline{x}$  on top of the stack by  $\underline{uvw}$ , Such that  $\underline{u}$  appears on top of the stack



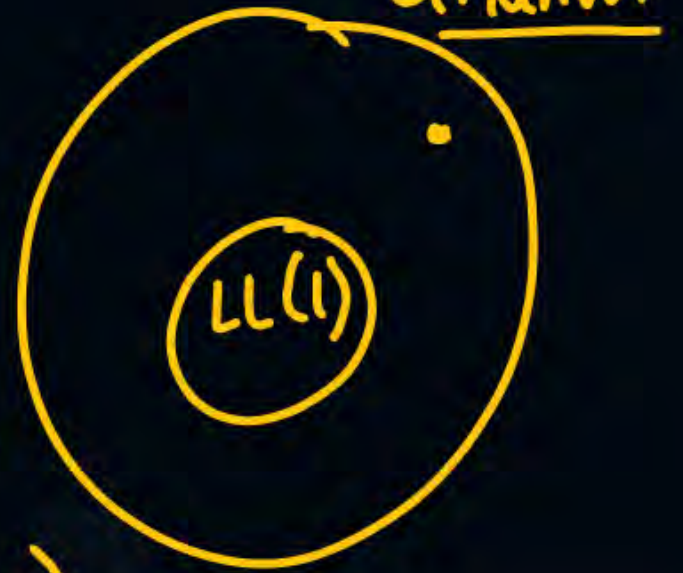
Given a grammar, is it  $LL(1)$  or not :-



→ If the  $LL(1)$  parsing table doesn't contain multiple entries, then such grammars are  $LL(1)$

→ A grammar will definitely not be  $LL(1)$  if it is ambiguous, or has left recursion or if it is non deterministic (common prefix)

→ Not all grammars that are unambiguous and free from left recursion and non determinism are  $LL(1)$





Rule 1:- A grammar without  $\epsilon$  rules is LL(1)

if each production is of form

$$A \rightarrow \alpha_1 / \alpha_2 / \alpha_3 \dots / \alpha_n$$

$\text{First}(\alpha_1)$ ,  $\text{First}(\alpha_2)$  ...  $\text{First}(\alpha_n)$  are mutually exclusive

(Every LL(1) grammar is unambiguous)





Rule 2 - A grammar with ' $\epsilon$ ' productions

is LL(1) if for each production of the form

$A \rightarrow \alpha / \epsilon$ ,  $\text{First}(\alpha)$  and  $\text{Follow}(A)$  must be mutually exclusive





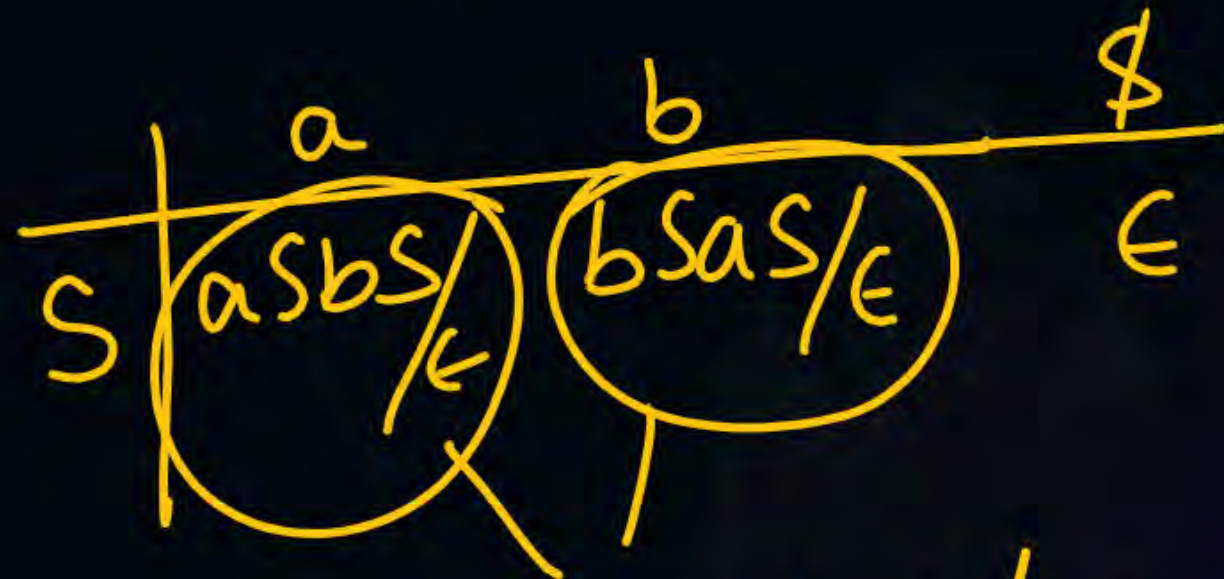
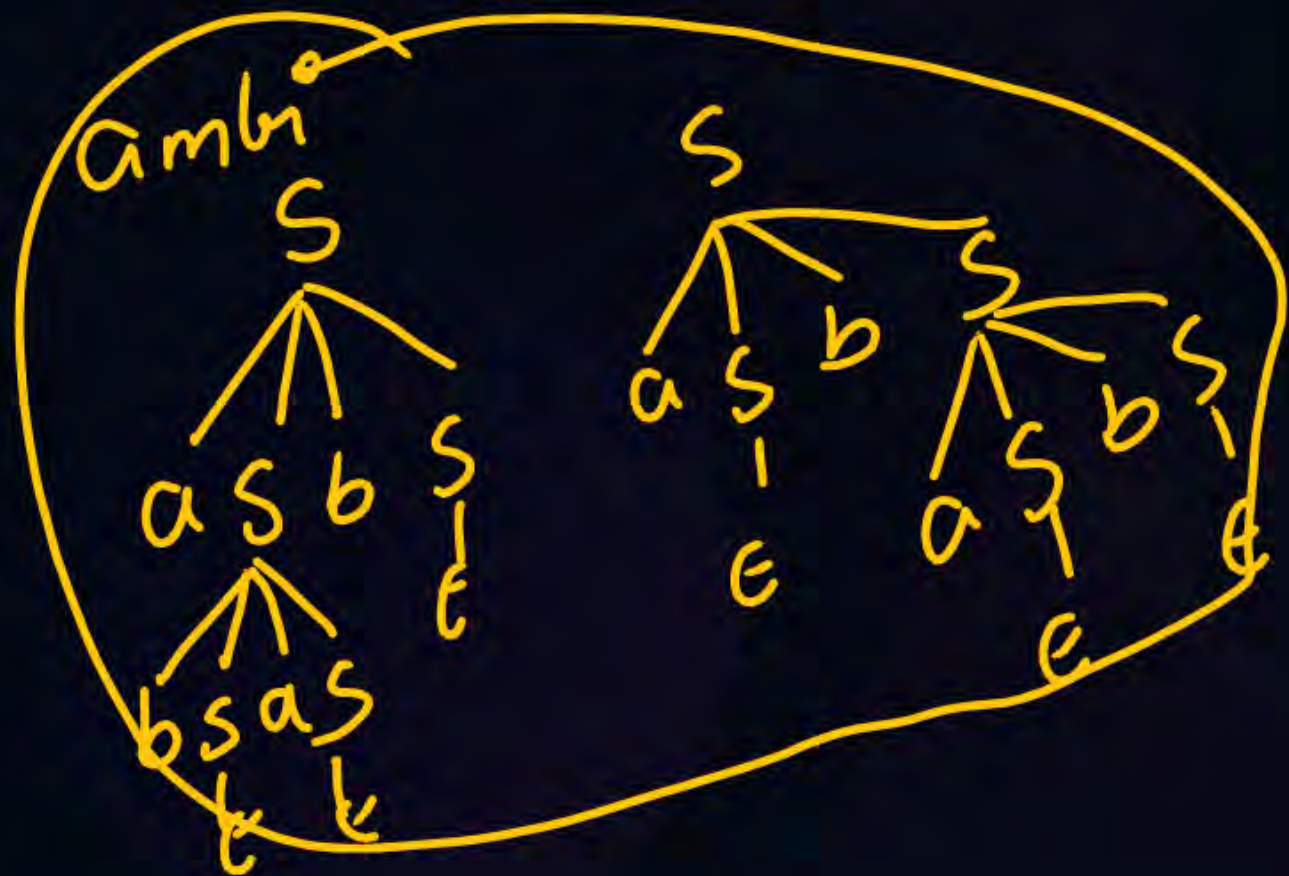
check whether following grammars are LL(1) or not:-

$S \rightarrow aSbS$   
 $S \rightarrow bSaS$   
 $S \rightarrow \epsilon$

NOT LL(1)

How many questions?

a) 10 b) 20



multiple entries



Thank you gourish Ghai

$S \rightarrow aABb$  {a}

$A \rightarrow c/\epsilon \rightarrow \{d, b\} \cap \{\epsilon\}$

$B \rightarrow d/\epsilon \rightarrow \{d\} \cap \{b\}$

a) LL(1) b) not LL(1)

|   | a                    | b                        | c                 | d                        | \$ |
|---|----------------------|--------------------------|-------------------|--------------------------|----|
| S | $S \rightarrow aABb$ |                          |                   |                          |    |
| A |                      | $A \rightarrow \epsilon$ | $A \rightarrow c$ | $A \rightarrow b$        |    |
| B |                      | $B \rightarrow d$        |                   | $B \rightarrow \epsilon$ |    |



$S \rightarrow \check{A} / (a)$   
 $A \rightarrow a$

|   | a                            | \$                |
|---|------------------------------|-------------------|
| S | $S \rightarrow A$<br>$\cdot$ | $S \rightarrow a$ |
| A |                              |                   |

↓ multiple entries  
 not LL(1)

a) LL(1)    b) Not LL(1)





$$S \rightarrow \underline{a} B / (\epsilon)$$

$$B \rightarrow \underline{b} C / (\epsilon)$$

$$C \rightarrow \underline{c} S / (\epsilon)$$

|   | a                  | b                  | c                  | \$                       |
|---|--------------------|--------------------|--------------------|--------------------------|
| S | $S \rightarrow aB$ |                    |                    | $S \rightarrow \epsilon$ |
| B |                    | $B \rightarrow bC$ |                    | $B \rightarrow \epsilon$ |
| C |                    |                    | $C \rightarrow cS$ | $C \rightarrow \epsilon$ |

a) LL(1)

b) Not LL(1)

$$\text{follow}(S) = \{ \$ \}$$

$$\text{follow}(B) = \{ \$ \}$$

$$\text{follow}(C) = \{ \$ \}$$

LL(1) ✓



$S \rightarrow \cancel{A.B}$   
 $A \rightarrow \underline{a}/\cancel{E}$   
 $B \rightarrow \underline{b}/\cancel{E}$

|   | a                  | b                  | \$                 |
|---|--------------------|--------------------|--------------------|
| S | $S \rightarrow AB$ | $S \rightarrow AB$ | $S \rightarrow AB$ |
| A | $A \rightarrow a$  | $A \rightarrow E$  | $A \rightarrow E$  |
| B |                    | $B \rightarrow b$  | $B \rightarrow E$  |

a) LL(1)    b) not LL(1)

LL(1)





$S \rightarrow aSA / \epsilon$   
 $A \rightarrow c / \epsilon$

a) LL(1)

b) Not LL(1)

$\text{Follow}(S) = \{ \$, c \}$

|   | a                   | c                        | \$                       |
|---|---------------------|--------------------------|--------------------------|
| S | $S \rightarrow aSA$ | $S \rightarrow \epsilon$ | $S \rightarrow \epsilon$ |
| A |                     | $A \rightarrow c$        | $A \rightarrow \epsilon$ |

X  $\text{LL}(1)$  ✓  
 Not



$S \rightarrow A^x$   
 $A \rightarrow \cancel{B}b / \cancel{C}d$   
 $B \rightarrow \underline{a}B / \underline{\epsilon}$   
 $C \rightarrow \underline{\cancel{C}}C / \underline{\epsilon}$

|   | a                  | b                        | c                  | d                        |
|---|--------------------|--------------------------|--------------------|--------------------------|
| S |                    |                          |                    |                          |
| A | $A \rightarrow Bb$ | $A \rightarrow Bb$       | $A \rightarrow cD$ | $A \rightarrow cD$       |
| B | $B \rightarrow aB$ | $B \rightarrow \epsilon$ |                    |                          |
| C |                    |                          | $C \rightarrow cC$ | $C \rightarrow \epsilon$ |
|   |                    |                          |                    |                          |

a) LL(1)

b) not LL(1)



$S \rightarrow aAa/\epsilon$   
 $A \rightarrow abS/\epsilon$

a) LL(1)

b) Not LL(1)

|   | a                                               | b | \$ |
|---|-------------------------------------------------|---|----|
| S | $S \rightarrow aAa$<br>$S \rightarrow \epsilon$ |   |    |
| A |                                                 |   |    |

Follow(S) = { \$, a }

Not LL(1)



$S \rightarrow iEt(S)S'/a$

$S' \rightarrow \epsilon S/\epsilon$

$E \rightarrow b$

a) LL(1)

b) Not LL(1)

Follow( $S'$ )  
 $= \text{Follow}(S) \epsilon$   
 $= \{\$ \} \cup \text{First}(S')$

|    | i                      | t | a                 | c                                                       | b |
|----|------------------------|---|-------------------|---------------------------------------------------------|---|
| S  | $S \rightarrow iEtSS'$ |   | $S \rightarrow a$ |                                                         |   |
| S' |                        |   |                   | $S \rightarrow \epsilon$<br>$S' \rightarrow \epsilon S$ |   |
| E  |                        |   |                   |                                                         |   |

Not LL(1)

multiple



## Recursive descent parser (Not important)



Writing recursive functions for productions is called  
Recursive descent parser.





$\checkmark E \rightarrow \checkmark i \checkmark E'$

$i + i (\$)$   
 $\uparrow \uparrow \uparrow \uparrow$

$E' \rightarrow \underline{+ i E'} / (E)$

$E()$

$\{ \text{if } (l == i)$   
 $\{ \underline{\text{match}(i)},$   
 $\underline{E'()},$   
 $\}$

$\times$   
 (RD)

$\times$   
 (LALR(1))

$E'()$

$\{ \text{if } (l == \underline{+})$   
 $\{ \underline{\text{match}(+' )},$   
 $\underline{\text{match}(+' )},$   
 $E'()$

$\}$   
 $\text{else return;}$   
 $E$

char l;

$l = \text{getchar}();$

$\text{match}(\text{char } t)$

$\{ \text{if } (l == t) \checkmark$

$l = \text{getchar}();$

else

$\text{printf}(\text{"error"});$

$\}$   
 $\text{main}()$

$\{$

$\underline{E()};$

$\text{if } (l \neq (\$)) \checkmark$

Parsing successful;

$\}$



**THANK - YOU**